

Prior art: who has built which piece

Component matrix for *critical Dicke electrometry*

Extending intracavity Rydberg-EIT microwave sensing (Peng *et al.*, JOSA B **35**, 2272) into the superradiant phase transition (Zhang *et al.*, PRL **110**, 090402). August 20, 2026

Legend: ✓ present — absent —/✓ *the cell that decides the strand*. **CRT** = counter-rotating light–matter terms (their presence is what permits a $\mathbb{Z}_2$ superradiant transition). **MW** = **measurand** means the microwave field is the quantity being measured, not merely a drive.

Paper	EIT	Cav.	$g\sqrt{N}$	CRT	Transition	Ryd. int.	MW = meas.	Th/Exp	Headline result
A. Cavity + Rydberg electrometry everything but the counter-rotating terms $\Rightarrow$ no transition at all									
Peng 2018 JOSA B 35, 2272 ★	✓	✓	— weak	—	—	—	✓	Th	$8\times$ lower detectable field; $>20\times$ resolution
Yang 2020 JOSA B 37, 1664	✓	✓	✓ coll. Rabi	—	—	—	✓	Th	$7\times$ over plain EIT; four-peak transmission
Wang 2023 JOSA B 40, 2604	✓	✓	✓ strong	—	—	—	✓	Th	$196.7\times$ / $26.2\times$; floor 396.5 nV cm^{-1}
Jing 2020 Nat. Phys. 16, 911	✓	—	—	—	—	—	✓ superhet	Exp	$55 \text{ nV cm}^{-1} \text{ Hz}^{-1/2}$
Liu 2025 Chin. Phys. Lett. 42, 053201	✓	— MW cav.	—	—	—	—	✓	Exp	18 dB power sensitivity
arXiv:2502.20792 cavity superhet	✓	✓	✓	—	—	—	✓ superhet	Exp	19 dB sensitivity gain
B. Critical Rydberg electrometry has criticality <i>and</i> sensing — but the interaction-driven transition, not light–matter									
Ding 2022 Nat. Phys. 18, 1447	✓	—	—	—	✓ NEPT / bistability	✓ drives it	✓	Exp	$\text{FI} \times 10^3$ vs. independent particles; $49 \text{ nV cm}^{-1} \text{ Hz}^{-1/2}$
Wang 2025 arXiv:2502.19761	✓	✓	—	—	✓ NEPT / bistability	✓ drives it	✓	Exp	classical FI $\times 100$ from cavity; $2.6 \text{ nV cm}^{-1} \text{ Hz}^{-1/2}$
C. Dicke model / phase structure has the transition, the cavity and the interactions — and senses nothing									
Dimer 2007 / Baden 2014 PRA 75, 013804 / PRL 113, 020408	—	✓	✓	✓ Raman	✓ $\mathbb{Z}_2$ Dicke	—	—	Th / Exp	Engineered Dicke QPT via cavity-assisted Raman
Zhang 2013 PRL 110, 090402 ★	2-photon	✓	✓	— RWA	✓ $U(1)$ polariton + SRS	✓	—	Th (QMC)	Superradiant solid; first-order boundaries
arXiv:2503.13949 anisotropic engineering	—	✓	✓	✓ tunable $0 \rightarrow \infty$	✓	✓	—	Th	Periodic-driving recipe for the anisotropic Dicke model
arXiv:2511.22230 Dicke–Ising QMC	—	✓	✓	✓	✓ SR / SRS / Solid- $\frac{1}{2}$	✓	— MW = <i>drive</i>	Th (QMC)	Orders of all transitions; blockade suppresses cavity occupation
Garbe 20/22, Rams 18, Gietka 21/22, Gyhm 25	—	✓	✓	✓	✓	—	— generic param.	Th	Critical-metrology scaling — and its no-gos
D. This proposal the one combination nobody occupies									
Critical Dicke electrometry	✓	✓	✓	✓	✓ $\mathbb{Z}_2$ Dicke	optional	✓ sets λ_c	Th	$\lambda_c = \frac{1}{2}\sqrt{\omega\tilde{\omega}_0}$ is a function of the measurand

★ = the two seed papers this project starts from.

Full discussion in `dicke_rydberg_sensing.pdf`, §6. Reading notes and references overleaf.

Reading the matrix

Every individual column is occupied, and each strand is exactly *one column* short of the proposal.

- **Strand A** lacks counter-rotating terms, so there is no transition at all. It stops at a collective Rabi splitting read out spectroscopically.
- **Strand B** has criticality and sensing, but its critical point is set by atomic density and interaction strength: it drifts with temperature and stray fields, it is hysteretic (the discriminator depends on scan direction), and it cannot be tuned quickly.
- **Strand C** has the genuine $\mathbb{Z}_2$ Dicke transition and does no sensing. The near-miss is arXiv:2511.22230, where microwave fields are *in* the setup — but as the drive that engineers the model, never as the measured quantity.

The empty cell that defines the project is the intersection of the last two columns — a $\mathbb{Z}_2$ Dicke transition whose critical point *is* the measurand:

$$\lambda_c = \frac{1}{2}\sqrt{\omega\tilde{\omega}_0}, \quad \tilde{\omega}_0 = \delta + \Omega_\mu/2, \quad \Omega_\mu = dE_\mu/\hbar.$$

Two cells worth staring at

- **Zhang 2013** $\rightarrow$ **– (RWA)**. Their coupling $(g/\sqrt{N})\sum_i(b_i^\dagger a_i \psi + \text{H.c.})$ conserves excitation number — which is why their phase diagram is drawn against a chemical potential. Their “generalised Dicke model” is therefore of Tavis–Cummings type and their superradiance is a $U(1)$ polariton condensation, not the $\mathbb{Z}_2$ transition this proposal needs. *The seed paper is one column short of the required model, so the extension is real rather than a re-derivation.*
- **arXiv:2503.13949** $\rightarrow$ **✓ (tunable $0 \rightarrow \infty$)**. The enabling technology, already published — and it fills exactly the cell Zhang 2013 leaves empty, for the Rydberg platform specifically.

The same RWA restriction runs through the whole of strand A. **Counter-rotating terms are the single ingredient separating this proposal from that entire lineage.**

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